

Date Format Conversion

There are multiple ways to specify a date. You have several clients that wish to be able to convert from one format to another. You will solve the first portion of this problem by completing the homeworks and projects up to and including Project 3. For Project 5, we will complete the solution.

Here is some background information. The method to specify a date used almost exclusively in the United States is referred to as the **MDY** format. This **MDY** format is simply a numeric sequence for Month/Day/Year. For example, the date for Exam #1 for our class would be shown as 9/23/2025. The month and day will be represented by one or two digits, and the year will be represented by four digits.

Another method is the **Ordinal Date** (*Often Mistakenly Called a Julian Date*)

This is the more common method used in business, manufacturing, and programming and the purpose of this case is to convert dates written in **MDY** format to **Ordinal Date** format.

- **What is Ordinal Date format?** A simple date format that represents a date as the year, followed by a dash and then the day number within that year (from 1 to 365, or 366 in a leap year) expressed as a three digit number.
- **Format:** It's expressed as YYYY-DDD
- **Example:**
 - January 1, 2025, would be 2025-001.
 - February 1, 2025, would be 2025-032.
 - The ordinal representation for the date of our first exam would be 2025-266 because that day is the 266th day of the year 2025.

Ordinal dates are used in many fields where simplicity, data processing, and removing ambiguity are critical. While a date like "July 10, 2025" is easy for people to read, the ordinal date 2025-191 is often much more useful for computers and record-keeping.

Here are some common, real-world examples where ordinal dates are used:

1. Manufacturing & Logistics

This is one of the most common uses. Ordinal dates are perfect for creating lot numbers and batch codes to track products.

- **Example:** You pick up a product with a lot code stamped on it: ****2025191-A42****
- **Meaning:** This instantly tells the manufacturer that the product was from the first batch (A) produced on the **191st** day of **2025** (which is July 10, 2025). If there's a recall, they can trace the exact production run with no confusion.

2. Software & Data Management

Programmers and system administrators frequently use ordinal dates for naming files because it's simple and sorts correctly.

- **Example:** Automated system logs or database backups might be named:
 - system_log_2025191.txt
 - backup_mainDB_2025191.zip
- **Why it's used:** This format avoids slashes (/) which are illegal in filenames, and it ensures files list in perfect chronological order when sorted alphabetically.

3. Military & Government

The U.S. military and various government agencies (like NASA) have long used this format for its clarity. In military circles, this is often what they mean when they say "Julian Date."

- **Example:** A supply request or operational plan might schedule a delivery for 2025-191.
- **Why it's used:** It removes international confusion. A date like 10/7/25 could mean October 7th **or** July 10th depending on the country. The 2025-191 format is unambiguous to everyone involved.

4. Food & Beverage Industry

Similar to manufacturing, ordinal dates are often used for "packed on" or "canned on" dates. You may have seen these on cans or bottles without realizing what they were.

- **Example:** A can of soda might have a code like P2025191 printed on the bottom.

To illustrate the first problem that will need to be solved, let's look at an example.

Question: What is the Ordinal Date for the MDY date: 3/1/ 2025?

Answer: 2025-060.

Question: What about 3/1/2024?

Answer: 2026-061. Why because 2024 is a leap year, so February had 29 days.

So, it seems that in order to make this conversion, one of the things that we need to know is whether or not the year in question was a leap year. That is the task of the first part of this case. Homework 5 and projects 1, 2, and 3 are all focused only on this.

Here are the rules for determining what years are leap years:

- Primary rule: The year must be evenly divisible by **4**. For example, 2024 was a leap year ($2024 / 4 = 506$ with no remainder), but 2025 is not a leap year ($2025 / 4 = 506$ with remainder of 1).
- Exception to the primary rule: If that year is **also** evenly divisible by **100**, it is **not** a leap year. For example, **1900** was **not** a leap year because although it was evenly divisible by 4, it was also evenly divisible by 100.
- Exception to the exception. If a year is evenly divisible by 100 and also evenly divisible by **400**, then it **is** a leap year. For example, **2000** was a leap year because although it is evenly divisible 100, it is also evenly divisible by 400.

The task for Homework 5 and Projects 1, 2, and 3 is to input a date in MDY format and determine whether the year of that date is a leap year or not. Some summary outputs for this task are also required. ***At this point in time, we are NOT doing the full conversion to Ordinal Date. That will take place in Project 5. Additional information for Project 5 requirements will be provided when that assignment is posted.***

What follows applies to Homework 5, and Projects 1, 2, and 3.

The initial input to the program is the client number. Following that, a prompt and response method will be used to input the dates of interest in MDY format. Inputs come from the keyboard.

After each date is entered, the program will immediately output the year and whether or not that year is leap year. This output will take the form of the word TRUE or FALSE.

Following the last date to be input, the value of N for the response variable will signal that input is complete and that summary values should be output now. The summary outputs are the client number, a count of how many input dates were leap years, a count of how many input dates were not leap years, the percentage of dates entered that were leap years, and finally the year of the most recent leap year in this data set. Outputs are sent to the screen.

A test data set is provided for you to test your models.
